Climate Downcasting via Frequency-Domain Autoregressive Transformer Modelling

Goal: Dwncasting of climate predictions by earth system models to higher resolution

Setup: Train model on global reanalysis data

Pipeline:

Input: Weather data (here: wind speed) with equiangular resolution of shape (N_lat, N_lon)

Spherical Harmonic transform

Complex coefficients in lower triangular matrix of shape (N_lat, N_lat)

Flattening, element-wise standardization, torch.view_as_real

Coefficients fo shape (L,2) where L = triangular_number(N_lat)

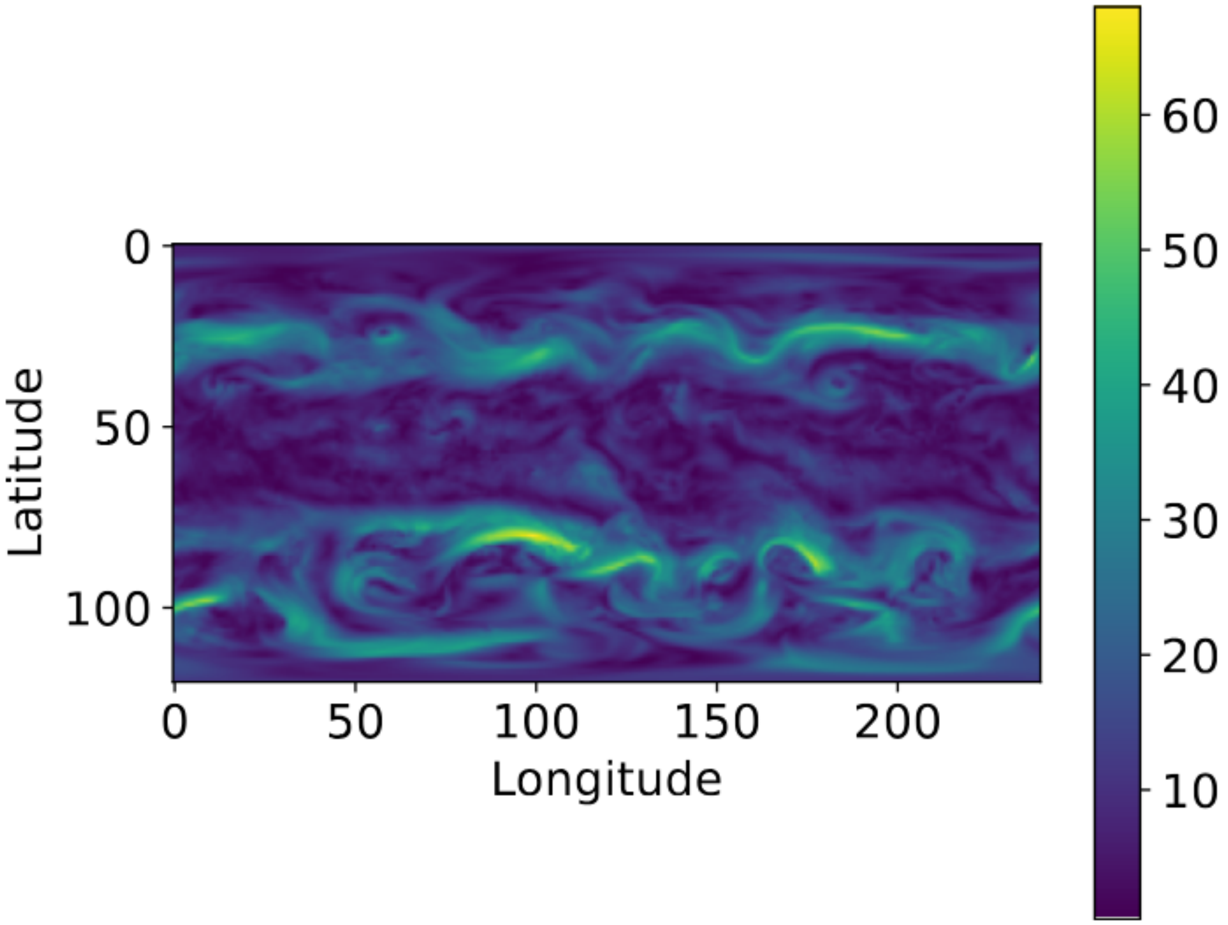
Transformer model that predicts distribution of next coefficient by parametrization as gaussian mixture model.
Autoregressively predict next coefficients starting at L,
ending at T = triangular_number(2*N_lat)

Model output of shape (T,2)

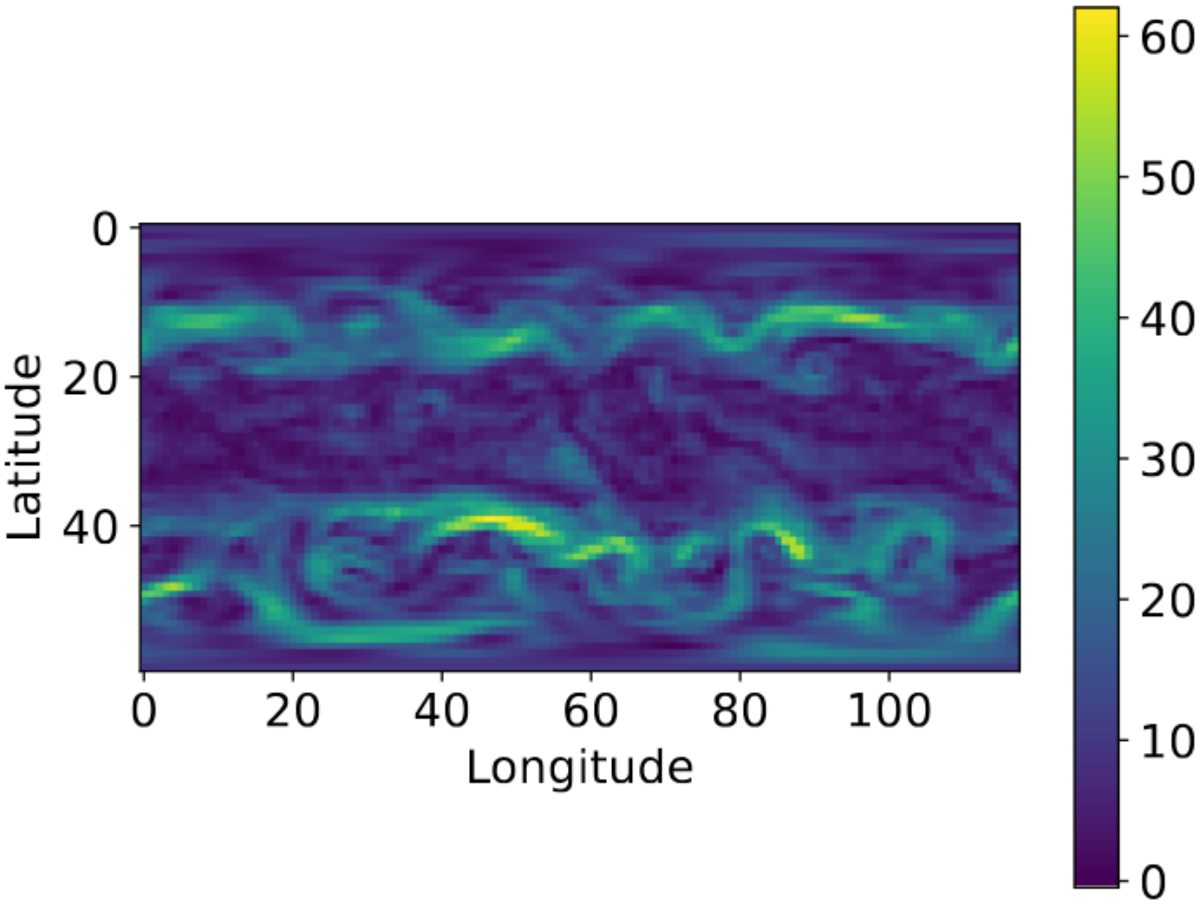
torch.view_as_complex, un-standardization, unflattening, inverse SH transform

Model prediction of shape (2*N_lat, 2*N_lon)

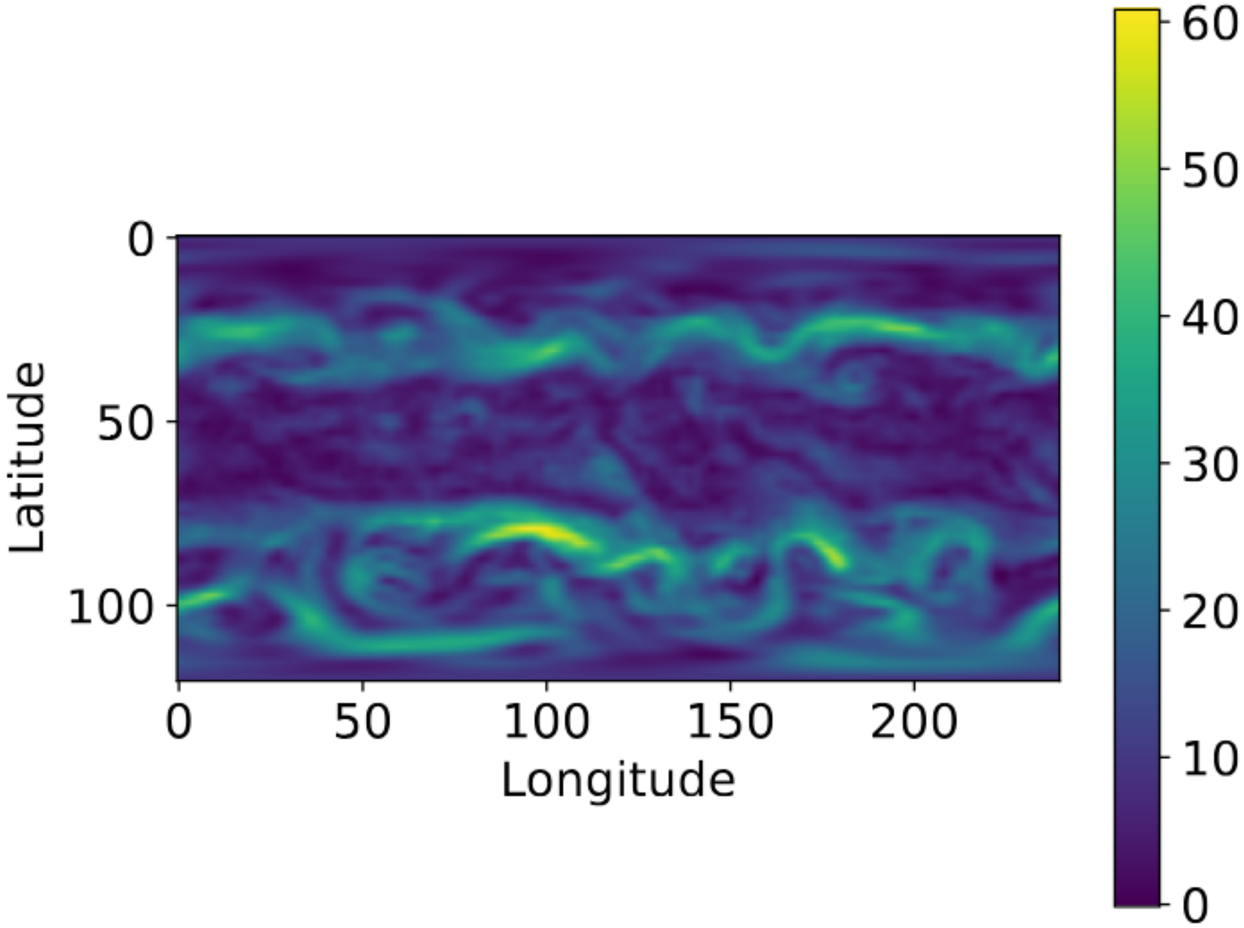
Ground truth data:



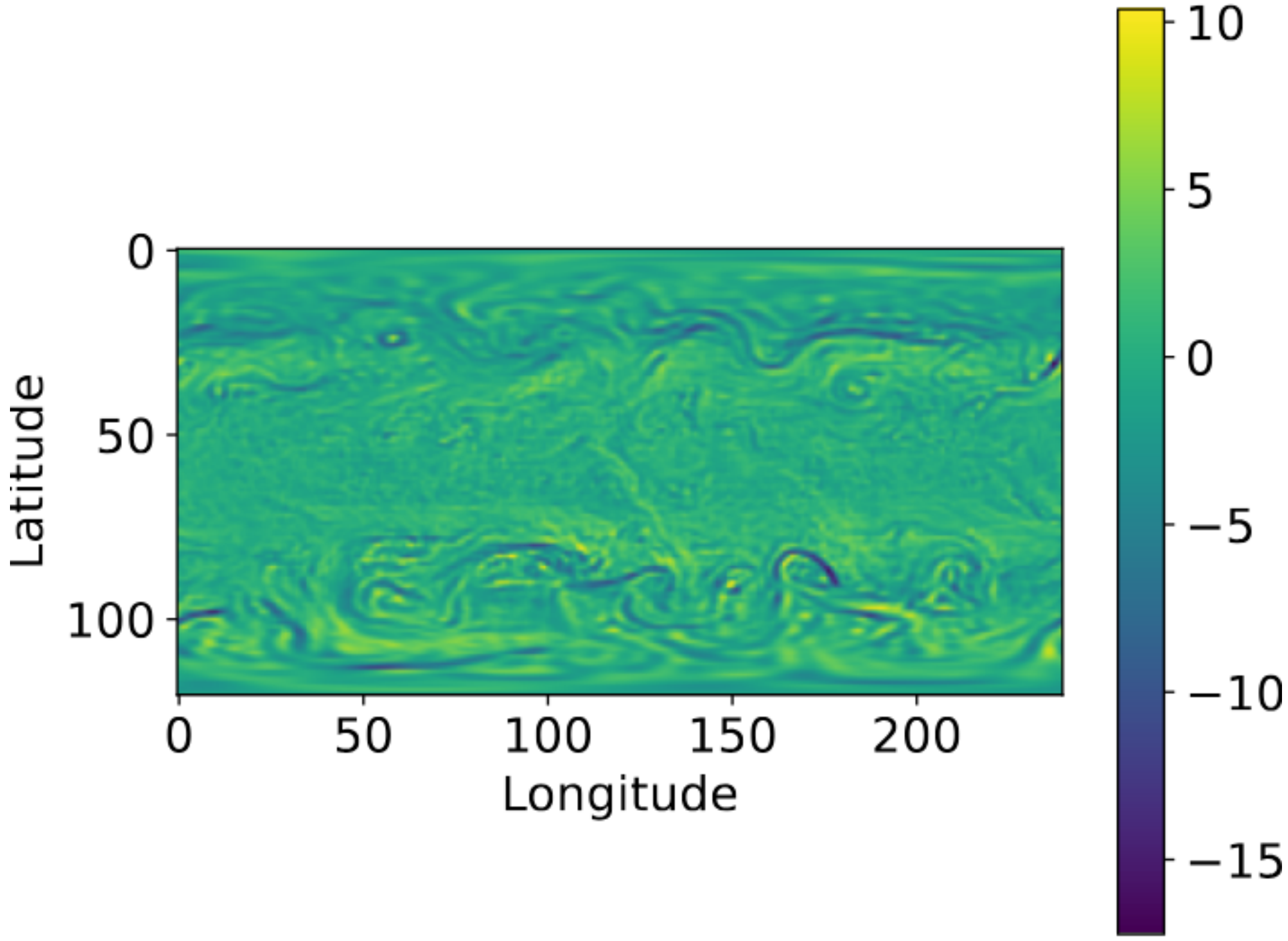
Model input:



Bilinear Interpoation:

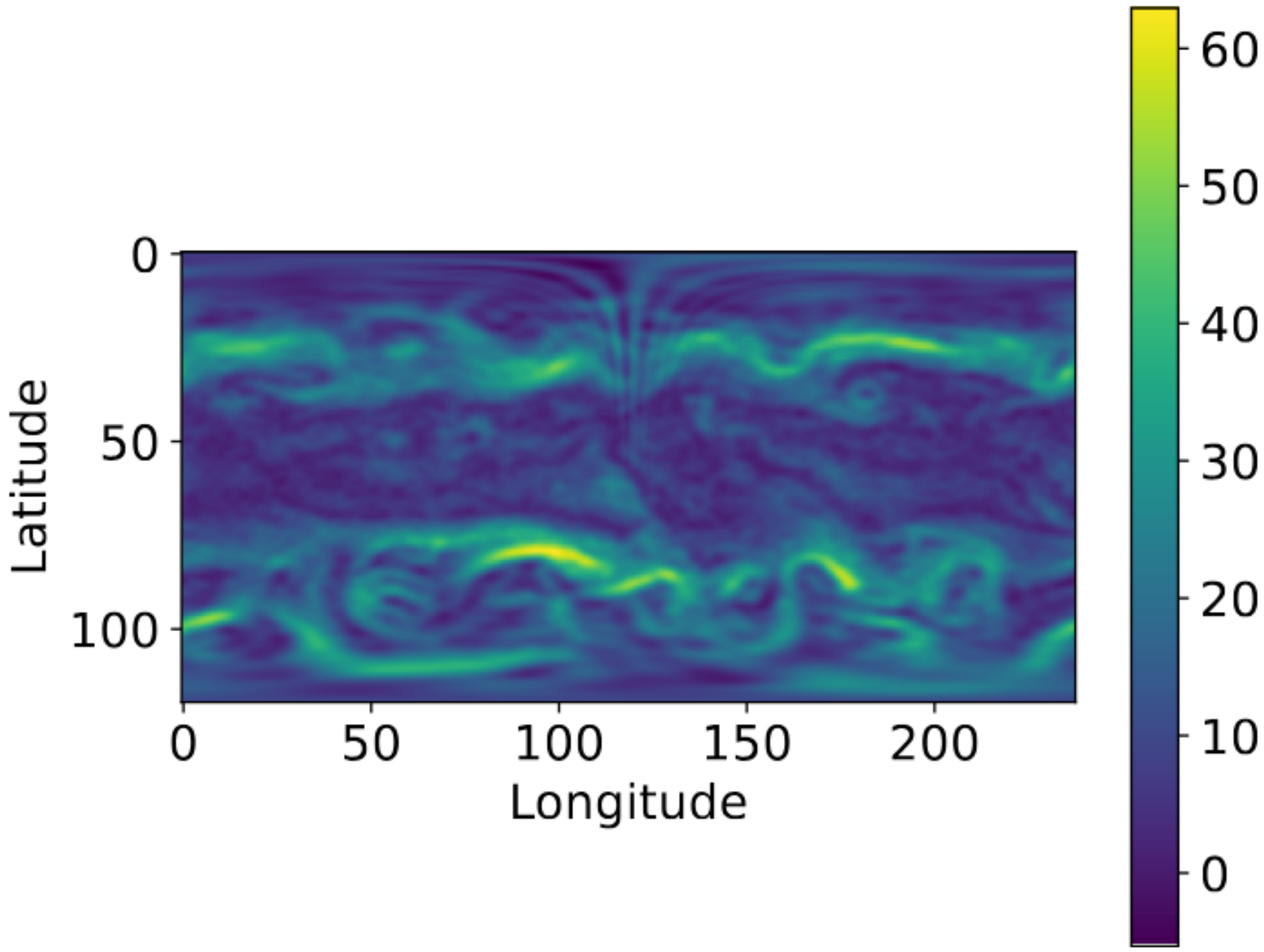


Difference to ground truth:

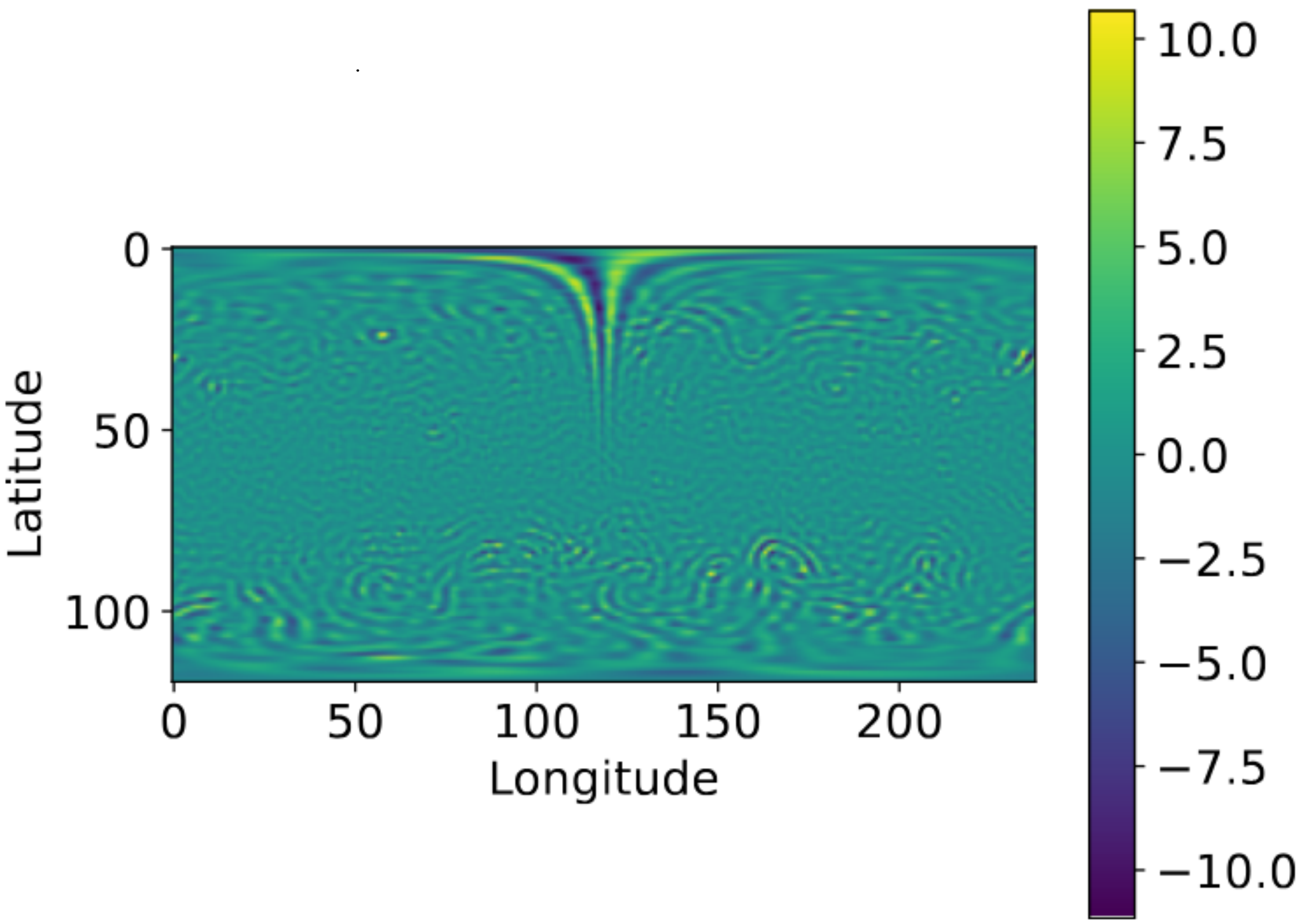


RMSE: 2.12 m/s

Transformer model: **16.3M parameters**



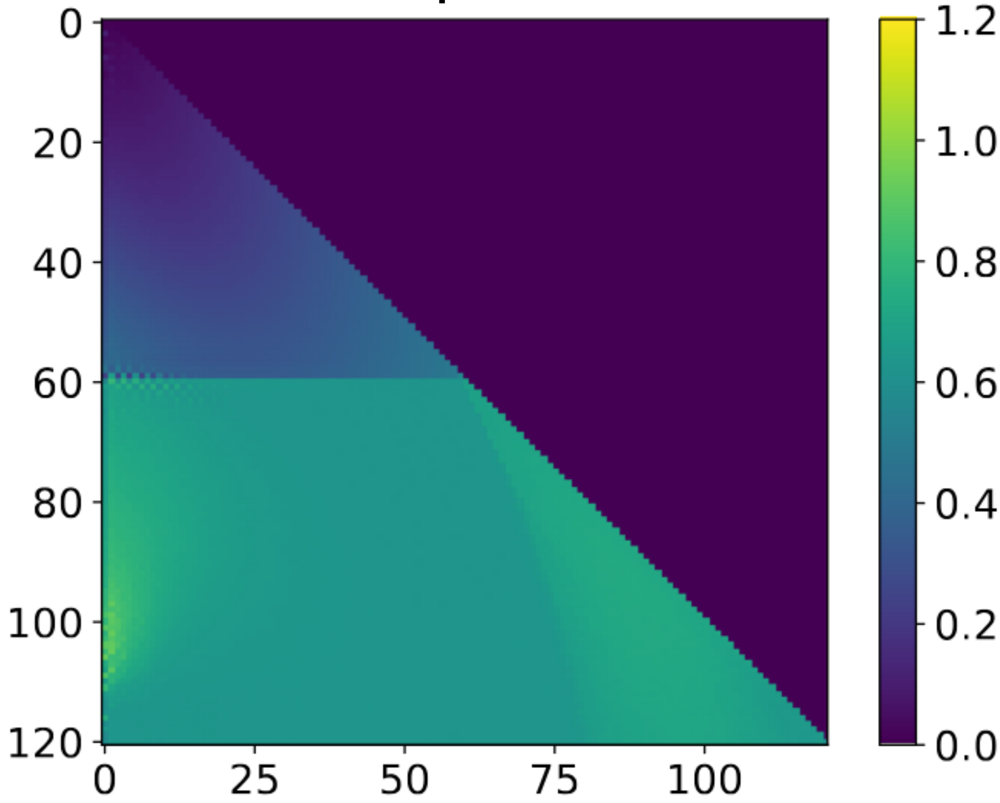
Difference to ground truth:



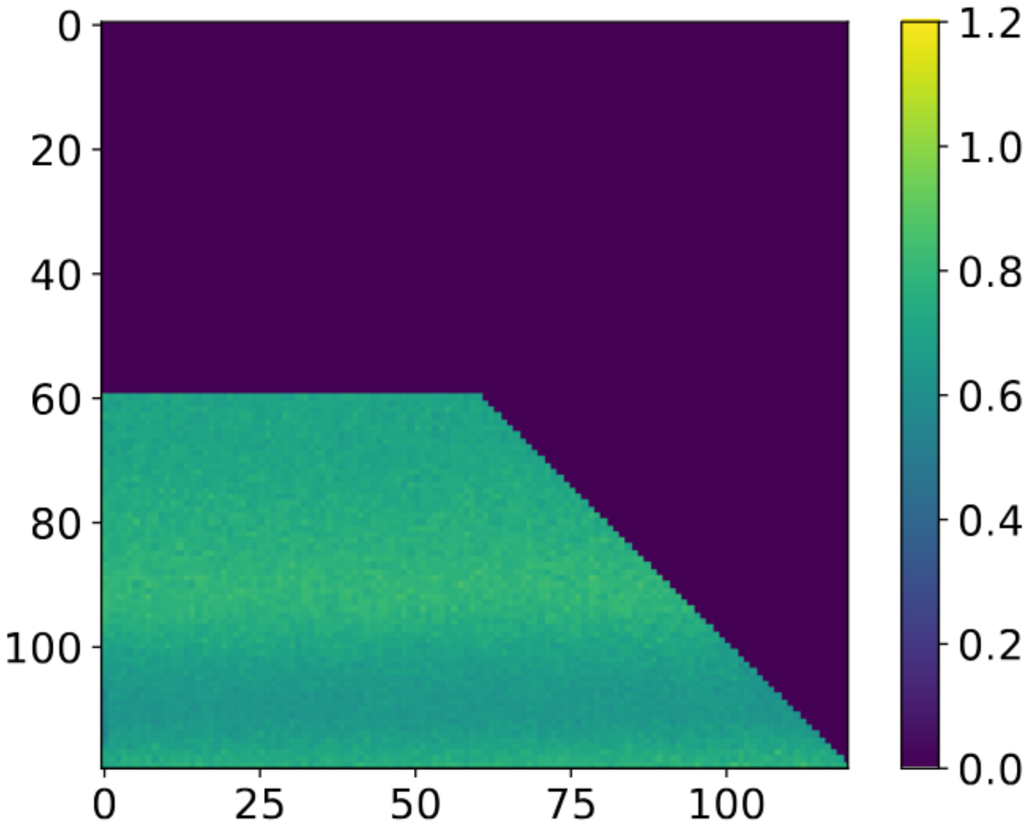
RMSE: 1.76 m/s

Coefficient RMSE Plots:

Interpolation



Transformer



=> Predictions don't seem to get significantly worse in later autoregression steps

Next steps:

- **Find out where north pole artifact comes from**
- **Use KV-Caching to make inference faster**
- **Train larger model**